ELEC2104/9704 Tutorial 5 Solutions (2025)

1 Question 1

Find the Q-point for the diode in the figure below using

(a) the ideal diode model,

(b) the constant voltage drop model with $V_{on} = 0.6\,\text{V}$,

(c) Discuss the results. Which answer do you feel is most correct?,

(d) Use iterative analysis to find the actual Q-point if $I_S = 1\,\text{fA}$ and assume thermal voltage is $25\,\text{mV}$.

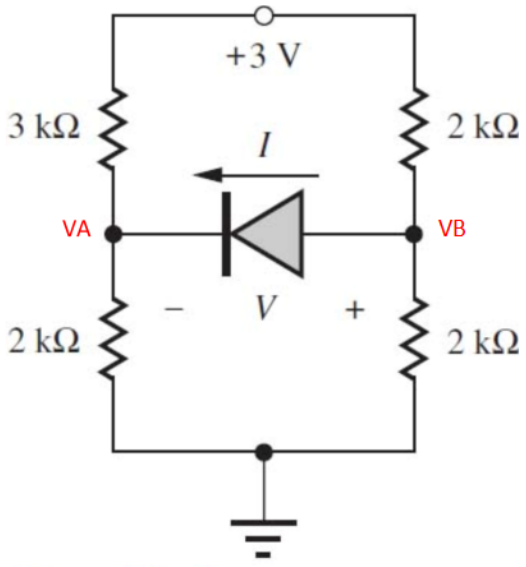


Figure 1: Q1 circuit diagram

Solution:

First, let's simplify the circuit using Thévenin equivalent. We'll apply the Thévenin theorem to both the left and right sides of the diode separately.

Step 1: Identify the portions to simplify
We'll find the Thévenin equivalent for the circuit to the left of the diode (looking from point V_A) and to the right of the diode (looking from point V_B).

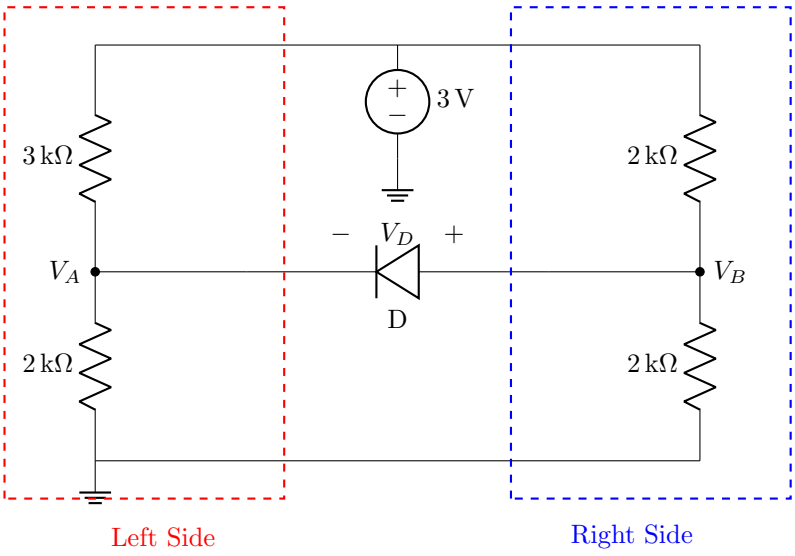


Figure 2: Original circuit with sections to be replaced by Thévenin equivalents

Step 2: Finding Thévenin Equivalent for Left Side
To find the Thévenin equivalent for the left side (looking from point V_A):

2a. Find $V_{th,left}$ (Open-circuit voltage at V_A):
Remove the diode (open circuit) and find the voltage at V_A using voltage divider:

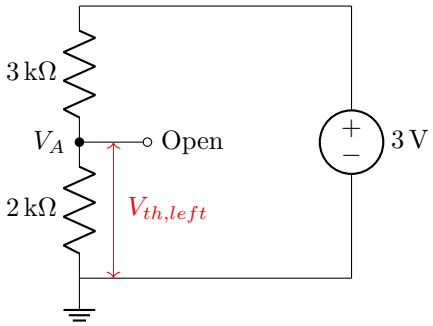


Figure 3: Finding $V_{th,left}$ - voltage divider with 2 kΩ and 3 kΩ

Using voltage divider: $V_{th,left} = 3\,\text{V} \times \frac{2\,\text{k}\Omega}{3\,\text{k}\Omega+2\,\text{k}\Omega} = 3 \times \frac{2}{5} = 1.2\,\text{V}$.

2b. Find $R_{th,left}$ (Thévenin resistance looking into V_A):
Replace voltage source with short circuit and find equivalent resistance:

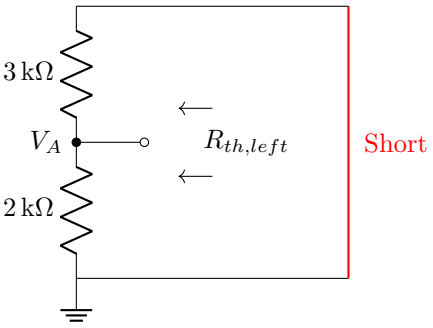


Figure 4: Finding $R_{th,left} = 2\,\text{k}\Omega \parallel 3\,\text{k}\Omega$

When the voltage source is shorted, the top and bottom rails are at the same potential (both connected to ground through the short). This means the $2\,\text{k}\Omega$ and $3\,\text{k}\Omega$ resistors are in parallel. We can redraw this more clearly:

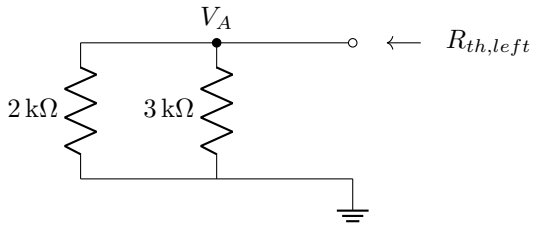


Figure 5: Equivalent parallel resistor configuration after shorting the voltage source

$R_{th,left} = 2\,\text{k}\Omega \parallel 3\,\text{k}\Omega = \frac{2\times3}{2+3}\,\text{k}\Omega = 1.2\,\text{k}\Omega$

Step 3: Finding Thévenin Equivalent for Right Side
Similarly for the right side (looking from point V_B):

3a. Find $V_{th,right}$ (Open-circuit voltage at V_B):

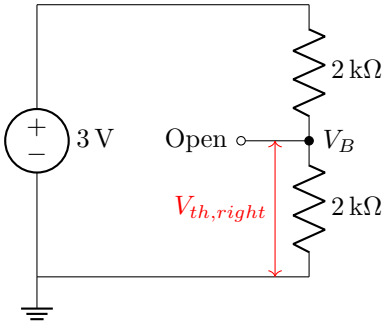


Figure 6: Finding $V_{th,right}$ - voltage divider with equal resistors

Using voltage divider: $V_{th,right} = 3\,\text{V} \times \frac{2\,\text{k}\Omega}{2\,\text{k}\Omega+2\,\text{k}\Omega} = 3 \times \frac{1}{2} = 1.5\,\text{V}$

3b. Find $R_{th,right}$ (Thévenin resistance looking into V_B):

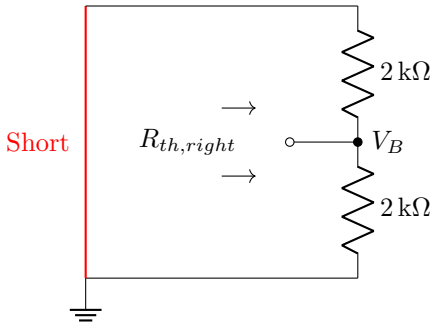


Figure 7: Finding $R_{th,right} = 2\,\text{k}\Omega \parallel 2\,\text{k}\Omega$

When the voltage source is shorted, the top and bottom rails are at the same potential. The two $2\,\text{k}\Omega$ resistors are in parallel. We can redraw this more clearly:

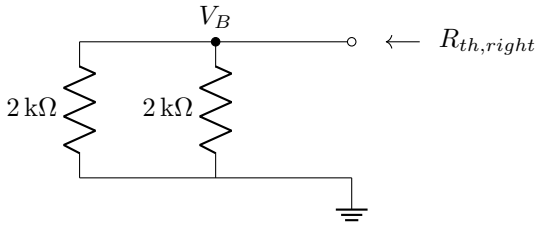


Figure 8: Equivalent parallel resistor configuration after shorting the voltage source

$R_{th,right} = 2\,\text{k}\Omega \parallel 2\,\text{k}\Omega = \frac{2\times 2}{2+2}\,\text{k}\Omega = 1\,\text{k}\Omega$

Step 4: First Simplified Circuit with Both Thévenin Equivalents
Now we can replace both sides with their Thévenin equivalents:

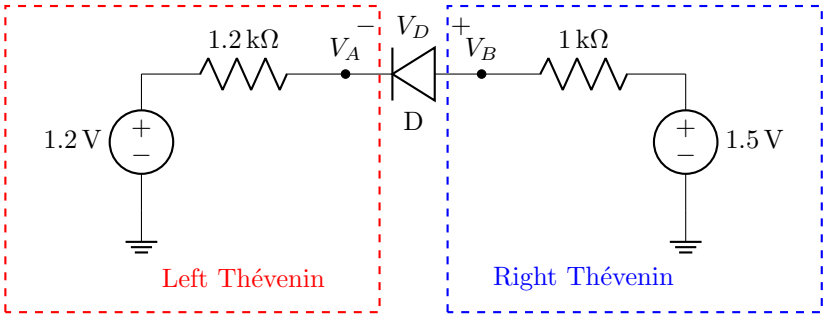


Figure 9: Circuit with Thévenin equivalents on both sides

Where the values of the Thévenin voltage sources and the resistors come from the previous steps:

$$V_{th,left} = 3\,\text{V} \times \frac{2\,\text{k}\Omega}{3\,\text{k}\Omega + 2\,\text{k}\Omega} = 1.2\,\text{V} \tag{1}$$

$$V_{th,right} = 3\,\text{V} \times \frac{2\,\text{k}\Omega}{2\,\text{k}\Omega + 2\,\text{k}\Omega} = 1.5\,\text{V} \tag{2}$$

$$R_{th,left} = 2\,\text{k}\Omega \parallel 3\,\text{k}\Omega = 1.2\,\text{k}\Omega \tag{3}$$

$$R_{th,right} = 2\,\text{k}\Omega \parallel 2\,\text{k}\Omega = 1\,\text{k}\Omega \tag{4}$$

Step 5: Further Simplification - Combining Sources and Resistors
We can further simplify by combining the voltage sources and resistors in series:

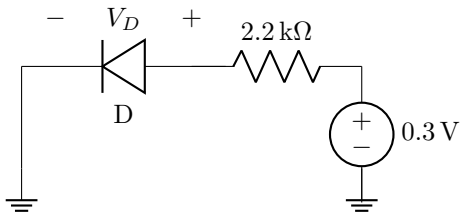


Figure 10: Final simplified circuit with combined Thévenin equivalent

The parameters are determined as:

$$V_{th} = V_{th,right} - V_{th,left} = 0.3\,\text{V} \tag{5}$$

$$R_{th} = R_{th,left} + R_{th,right} = 2.2\,\text{k}\Omega \tag{6}$$

1.1 Part A: Ideal Diode Model

Rules in Ideal Diode Models:

$$\begin{cases} \text{ON: Diode in forward bias } (I_D > 0, V_D = 0): & \text{diode } \approx \text{ short circuit,} \\ \text{OFF: Diode in reverse bias } (I_D = 0, V_D < 0): & \text{diode } \approx \text{ open circuit.} \end{cases}$$

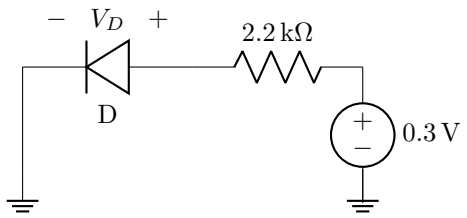


Figure 11: Thévenin equivalent circuit

By inspecting the Thévenin equivalent circuit found in the steps above (see Figure 11), we can see that the voltage source tries to forward bias the diode (positive terminal connected to anode). Since this would result in forward current, we assume that the diode is ON.

When the diode is ON in the ideal diode model, it acts as a short circuit ($V_D = 0$). We can redraw the circuit with this assumption:

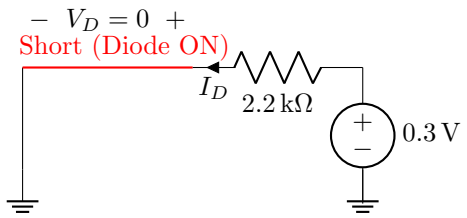


Figure 12: Thévenin equivalent circuit with ideal diode ON (shorted)

Now we can calculate the current through the circuit using Ohm's law:

$$I_D = \frac{V_{th}}{R_{th}} = \frac{0.3\,\text{V}}{2.2\,\text{k}\Omega} = 0.136\,\text{mA} \tag{7}$$

Since $I_D > 0$, our assumption that the diode is ON is correct. Therefore, the Q-point is:

$$(I_D, V_D) = (0.136\,\text{mA},\, 0\,\text{V}) \tag{8}$$

1.2 Part B: Constant Voltage Drop Model

Rules in Constant Voltage Drop Model:

$$\begin{cases} \text{ON: If } V_D \geq V_{D,on}: & V_D = V_{D,on},\ I_D > 0 \\ \text{OFF: If } V_D < V_{D,on}: & V_D < V_{D,on},\ I_D = 0 \end{cases}$$

where $V_{D,on} = 0.6\,\text{V}$ is the turn-on voltage.

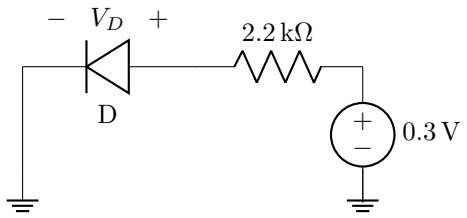


Figure 13: Thévenin equivalent circuit for constant voltage drop analysis

To determine if the diode is ON or OFF, we use the "assume and verify" approach. Since the available voltage (0.3V) is less than the turn-on voltage (0.6V), we start by assuming the diode is OFF.

When the diode is OFF (open circuit), no current flows through the circuit, so the voltage across the diode equals the full Thévenin voltage:

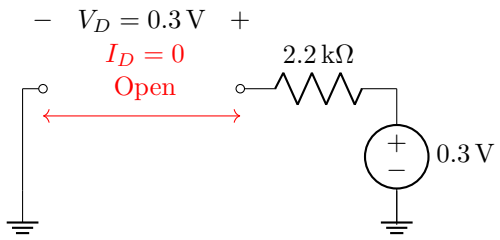


Figure 14: Circuit assuming diode is OFF (open circuit)

Since $I_D = 0\,\text{A}$ (diode is OFF), by Ohm's law, the voltage drop across the resistor is:

$$V_R = I_D \times R_{th} = 0\,\text{A} \times 2.2\,\text{k}\Omega = 0\,\text{V} \tag{9}$$

Therefore: $V_D = V_{th} - V_R = 0.3\,\text{V} - 0\,\text{V} = 0.3\,\text{V}$.
So, when the diode is OFF, we have $V_D = V_{th} = 0.3\,\text{V}$.

Since $V_D = 0.3\,\text{V} < V_{D,on} = 0.6\,\text{V}$, our assumption is correct: the diode is OFF.
Therefore, the Q-point is:

$$(I_D, V_D) = (0\,\text{A},\, 0.3\,\text{V}) \tag{10}$$

1.3 Part C: Which answer do you feel is most correct?

To determine which model is more accurate, let's check our results using the exponential diode equation (which we know to be the most accurate) and compare with the two models above.

The exponential diode equation is:

$$I_D = I_S \left[\exp\left(\frac{V_D}{V_T}\right) - 1\right] \tag{11}$$

Using typical values: $I_S = 10^{-15}\,\text{A}$ and $V_T = 25\,\text{mV}$ (room temperature).

To test the validity of our Part B analysis, let's assume the diode voltage is $V_D = 0.3\,\text{V}$ (as predicted when the diode is OFF) and see what current the exponential model predicts:

$$I_D = 10^{-15}\left[\exp\left(\frac{0.3}{0.025}\right) - 1\right] \tag{12}$$

$$I_D = 10^{-15}\left[\exp(12) - 1\right] = 163\,\text{pA} \tag{13}$$

Note: This is not the exact solution of the exponential for this given circuit (which requires iterative methods in Part D), but rather a check of what current would flow if $V_D = 0.3\,\text{V}$.

Conclusion: The **constant voltage drop model (Part B)** is more correct.

The exponential model shows that at $V_D = 0.3\,\text{V}$, the current is only 163 pA - essentially negligible for practical purposes. This confirms that the diode is effectively OFF, supporting the Part B analysis that predicted zero current and 0.3V across the diode.

Next, in Part D, let's properly solve the exponential model for this circuit (without making the $V_D = 0.3\,\text{V}$ assumption) using an iterative method.

1.4 Part D: Use iterative analysis to find the actual Q-point

For the most accurate analysis, we use the full exponential diode model with $I_S = 1\,\text{fA}$ and $V_T = 25\,\text{mV}$.

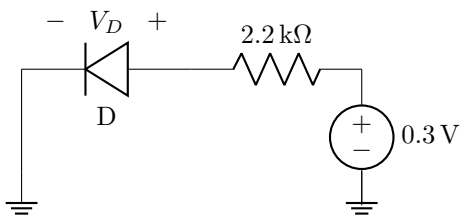


Figure 15: Thévenin equivalent circuit for iterative analysis

Step 1: Set up the system of equations
We have two constraints that must be satisfied simultaneously:
Diode equation:

$$I_D = I_S \left[\exp\left(\frac{V_D}{V_T}\right) - 1\right] = 10^{-15}\left[\exp\left(\frac{V_D}{0.025}\right) - 1\right] \tag{14}$$

Circuit equation (Ohm's law):

$$I_D = \frac{V_{th} - V_D}{R_{th}} = \frac{0.3 - V_D}{2200} \tag{15}$$

Step 2: Set up for Newton-Raphson iteration
We choose to iterate on current I_D rather than voltage V_D because iterating on current gives better numerical convergence for small currents. From (14), we can solve for V_D in terms of I_D:

$$V_D = V_T \ln\left(\frac{I_D}{I_S} + 1\right) = 0.025 \ln\left(10^{15} I_D + 1\right) \tag{16}$$

From (15), we can also solve for V_D:

$$V_D = V_{th} - R_{th} I_D = 0.3 - 2200 I_D \tag{17}$$

Since both expressions equal V_D, we set them equal:

$$0.025 \ln\left(10^{15} I_D + 1\right) = 0.3 - 2200 I_D \tag{18}$$

Rearranging to standard form $f(I_D) = 0$:

$$0.025 \ln\left(10^{15} I_D + 1\right) - 0.3 + 2200 I_D = 0 \tag{19}$$

This gives us our Newton-Raphson function:

$$f(I_D) = 0.025 \ln\left(10^{15} I_D + 1\right) + 2200 I_D - 0.3 \tag{20}$$

The derivative is:

$$f'(I_D) = \frac{0.025 \times 10^{15}}{10^{15} I_D + 1} + 2200 \tag{21}$$

Step 3: Newton-Raphson iteration
Starting with initial guess $I_{D0} = 10^{-15}\,\text{A}$ (i.e. the saturation current), we iterate:

$$I_{D(n+1)} = I_{D(n)} - \frac{f(I_{D(n)})}{f'(I_{D(n)})} \tag{22}$$

Step 4: Results
After convergence, we get:

$$I_D = 163\,\text{pA} \tag{23}$$

Finding V_D. Using (15) to find the corresponding voltage gives:

$$V_D = 0.3 - 2200 \times 163 \times 10^{-12} = 0.3 - 3.586 \times 10^{-7} = 0.300\,\text{V} \tag{24}$$

Therefore, the accurate Q-point is:

$$\boxed{(I_D, V_D) = (163\,\text{pA},\ 0.300\,\text{V})} \tag{25}$$

This confirms our Part C conclusion that the diode is essentially OFF, with negligible current flow.

Comparison of results:

- **Part A (Ideal model):** $I_D = 0.136\,\text{mA}$, $V_D = 0\,\text{V}$
- **Part B (Constant voltage drop):** $I_D = 0\,\text{A}$, $V_D = 0.3\,\text{V}$
- **Exponential model:** $I_D = 163\,\text{pA}$, $V_D = 0.300\,\text{V}$

Important: Loading the Circuit and why Thévenin Equivalent is Necessary

Common Misconception: Students often try to analyze this circuit without its Thévenin Equivalent by simply calculating V_A and V_B using voltage dividers, then finding $V_D = V_B - V_A$.

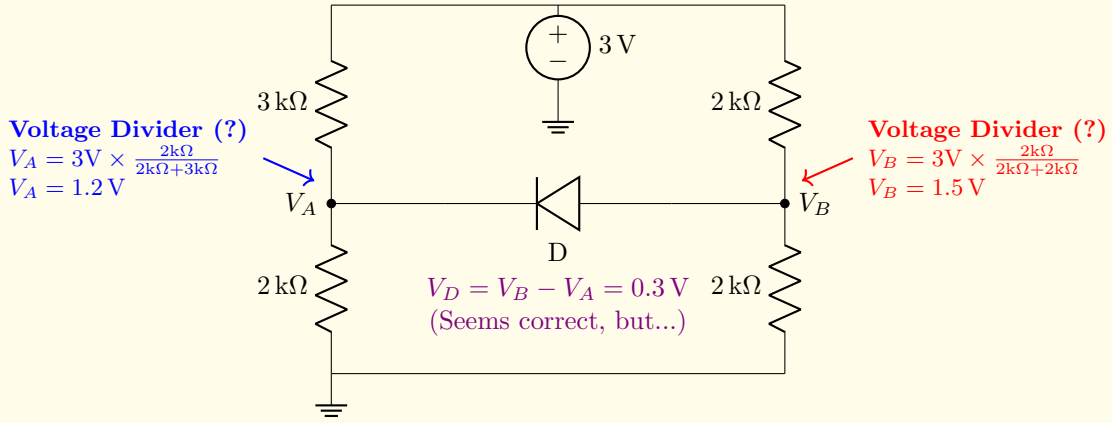


Figure 16: The tempting but flawed voltage divider approach

In this specific problem, this approach gives the correct answer (0.3V) because the diode is OFF with negligible current. However, this is **an incorrect solution!**

The Critical Issue - Loading the circuit when the Diode is ON:

- Assume a larger source voltage (e.g. 10 V) such that the diode turns ON
- When ON, the diode conducts significant current I_D
- This current flows through both 3kΩ/2kΩ and 2kΩ/2kΩ resistor networks, causing:
  - V_A to increase from its open-circuit value (current flows left into the node from the diode)
  - V_B to decrease from its open-circuit value (current flows right out of the node to the diode)

  Therefore, the resulting V_D is *less* than the subtraction of the voltage divider-derived values of V_B and V_A due to the current. This is called **loading** the circuit.
- The voltage divider method **only gives the correct answer when no current flows**. I.e., the circuit is *unloaded*.

Why Thévenin Equivalent Works:
The Thévenin Voltage (V_{th}) captures the open-circuit voltage (what the voltage dividers give you) if no current were flowing into the diode, while the Thévenin Resistance (R_{th}) captures *how this voltage changes with current through the diode.*
Example to Illustrate the Problem:
Consider if this circuit had a 10 V source instead of 3 V:

- Voltage divider approach gives $V_A = 4\,\text{V}$, $V_B = 5\,\text{V}$, incorrectly suggesting $V_D = 1\,\text{V}$
- Actual answer: Diode turns ON with $V_D = 0.645\,\text{V}$ and $I_D = 161\,\mu\text{A}$.

Try this example yourself by re-attempting Q1.d with a 10 V voltage source.

Contrast with MOSFET Circuits (Future Topic): Later in this course, you'll see MOSFET (transistor) circuits where we **do** use voltage dividers directly. The key difference is that MOSFET gates draw **zero DC current** ($I_G = 0$), so they don't load the voltage divider.

2 Question 2

The full-wave rectifier in the figure below is operating at a frequency of 60 Hz, and the rms value of the transformer output voltage is 18 V.

(a) What is the value of the dc output voltage if the diode voltage drop is 1 V?

(b) What is the minimum value of C required to maintain the ripple voltage to less than 0.25 V if $R = 0.5\,\Omega$?

(c) What is the PIV rating of the diode in this circuit?

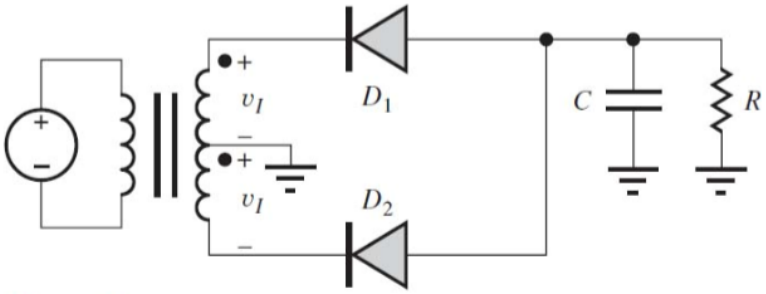


Figure 17: Q2 circuit diagram

Solution:

Looking at the circuit diagram, we can recognize that it is a full-wave rectifier with a center-tap transformer.
Understanding Current Flow:
This rectifier operates by utilizing both half-cycles of the AC input through two separate conduction paths:

- **Positive half-cycle of v_I:** The top terminal of the transformer secondary is positive. Current flows from the center-tap (ground), up through the load resistor R (from bottom to top), through diode D_1 (which is forward-biased), to the top terminal of the transformer. During this time, diode D_2 is reverse-biased and blocks current flow.
- **Negative half-cycle of v_I:** The bottom terminal of the transformer secondary is positive. Current flows from the center-tap (ground), up through the load resistor R (from bottom to top, same direction as positive half-cycle), through diode D_2 (which is forward-biased), to the bottom terminal of the transformer. During this time, diode D_1 is reverse-biased.

Key Insight: Notice that in both half-cycles, current flows through the load resistor in the *same direction* (from ground upward through the resistor). This unusual diode orientation forces current to flow FROM ground TO the DC output node, making V_{dc} negative relative to ground.

2.1 Part A

To find the DC output voltage, we need to trace the current path and account for the diode voltage drop.

Step 1: Determine peak voltage from given RMS voltage The peak transformer secondary voltage (from center-tap to either end) is:

$$V_p = V_{\text{rms}} \times \sqrt{2} = 18\sqrt{2} = 25.46\,\text{V}$$

This is just a conversion from the given $V_{rms} = 18$ V RMS voltage to peak voltage of a sine wave (AC).

Step 2: Account for diode voltage drop During each half-cycle, current flows through exactly one diode before reaching the load. Therefore:

$$V_{\text{dc,magnitude}} = V_p - V_{\text{on}} = 25.46 - 1 = 24.46\,\text{V}$$

Step 3: Determine polarity Looking at the circuit orientation and following the current flow:

- Current flows *from ground upward* to the V_{dc} node
- This means the V_{dc} node is at a *lower potential* than ground
- Therefore: $V_{\text{dc}} = -24.46\,\text{V}$

$$V_{dc} = -\left(V_p - V_{on}\right) = -\left(25.46 - 1\right) = -24.46\,\text{V} \tag{1}$$

Important Note: This circuit produces a *negative* DC output voltage because of the diode orientations. In a typical rectifier, current flows from V_{dc} to ground (positive output), but here the diodes are oriented to force current from ground toward the V_{dc} node.

2.2 Part B

We will apply the formulas in the red box in the table below.

Comparison of Rectifiers with Capacitive Filters

RECTIFIER PARAMETER	HALF-WAVE RECTIFIER	FULL-WAVE RECTIFIER	FULL-WAVE BRIDGE RECTIFIER
Filter capacitor	$C = \frac{V_P - V_{\text{on}}}{V_r}\frac{T}{R}$	$C = \frac{V_P - V_{\text{on}}}{V_r}\frac{T}{2R}$	$C = \frac{V_P - 2V_{\text{on}}}{V_r}\frac{T}{2R}$
PIV rating	$2V_P$	$2V_P$	V_P
Peak diode current (constant V_r)	Highest I_P	Reduced $\frac{I_P}{2}$	Reduced $\frac{I_P}{2}$
Surge Current	Highest	Reduced ($\propto C$)	Reduced ($\propto C$)

Figure 18: Comparison of Rectifiers with Capacitive Filters

Also, we can get the following parameters from the question.

$$\begin{aligned} \text{Operation frequency: } & f = 60\,\text{Hz}, \\ \text{Operation period: } & T = 1/f = 1/60\,\text{sec}, \\ \text{RMS transformer output: } & V_{\text{rms}} = 18\,\text{V}. \end{aligned} \tag{2}$$

We can determine the peak transformer output voltage by

$$V_p = V_{\text{rms}} \times \sqrt{2} = 18\sqrt{2}\,\text{V}. \tag{3}$$

Taking the formula of the required capacitance for the full-wave rectifier for voltage ripple $V_r = 0.25\,\text{V}$ and load resistance $R = 0.5\,\Omega$ (see Figure 18), we get:

$$C = \frac{V_p - V_{on}}{V_r} \cdot \frac{T}{2R} = \frac{18\sqrt{2} - 1}{0.25} \cdot \frac{1/60}{2 \cdot 0.5} = 1.63\,\text{F} \tag{4}$$

2.3 Part C

Taking the formula of the required PIV for the full-wave rectifier, we can get

$$PIV \geq 2V_p = 2 \cdot 18\sqrt{2} = 50.91\,\text{V} \tag{5}$$

3 Question 3

What are the dc output voltages V_1 and V_2 for the rectifier circuit in the figure below if $v_i = 40 \sin 377t$ and $C = 20{,}000\,\mu\text{F}$?

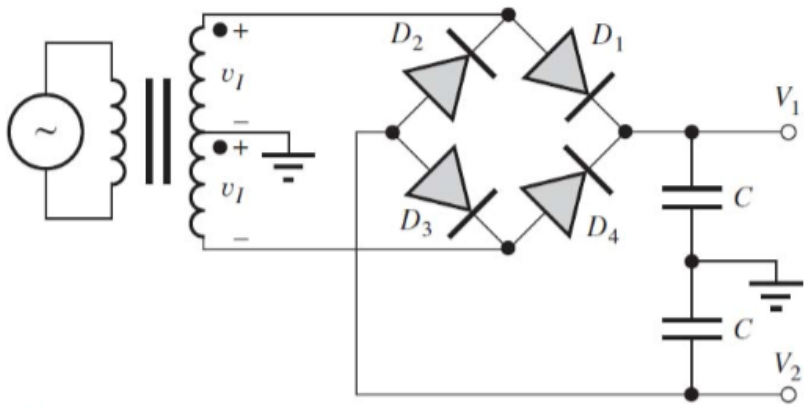


Figure 19: Q3 circuit diagram

Solution:

This circuit is a **dual-output center-tap rectifier** that produces both positive and negative DC voltages simultaneously.

Let the top and bottom capacitors be denoted C_1 and C_2, repectively.
Positive half-cycle of v_i:

- Upper terminal: $+40\,\text{V}$, Lower terminal: $-40\,\text{V}$
- D_1 forward-biased: Current flows via C_1 to ground, creating $V_1 > 0$
- D_3 forward-biased: Current flows from ground via C_2, creating $V_2 < 0$
- D_2 and D_4 are reverse-biased (OFF)

Negative half-cycle of v_i:

- Upper terminal: $-40\,\text{V}$, Lower terminal: $+40\,\text{V}$
- D_2 forward-biased: Current flows from ground via C_2, maintaining $V_2 < 0$
- D_4 forward-biased: Current flows via C_1 to ground, maintaining $V_1 > 0$
- D_1 and D_3 are reverse-biased (OFF)

From $v_i = 40 \sin 377t$, we can extract:

$$\begin{gathered}\text{Operation frequency: } \omega = 377\,\text{rad/s}, \\ \text{Peak voltage: } V_p = 40\,\text{V}\end{gathered} \tag{1}$$

For each output (V_1 and V_2), there is exactly one diode in the conduction path during each half-cycle. Using the diode turn-on voltage $V_{D,on} = 1\,\text{V}$ from Q2:

Voltage Calculations:

$$\begin{aligned} V_1 &= V_p - V_{D,on} = 40 - 1 = 39\,\text{V}, \\ V_2 &= -\left(V_p - V_{D,on}\right) = -(40 - 1) = -39\,\text{V}. \end{aligned} \tag{2}$$

Insights:

- V_1 is always positive because current always flows *from* V_1 *to* ground.
- V_2 is always negative because current flows *from* ground *to* V_2.
- Both outputs are full-wave rectified, providing continuous DC with minimal ripple
- The circuit effectively creates a dual-rail power supply ($+39\,\text{V}$ and $-39\,\text{V}$)